Chapter 8 Techniques of Integration

8.1 Basic Integration Techniques

Recall that if $F(x)$ is a differential function of x, and that $F'(x) = f(x)$, then

$$\int f(x)dx = F(x) + C.$$

Recall also that if we let $u = g(x)$, then

$$\int f(g(x))g'(x)dx = \int f(u)du.$$

TABLE 8.1 Basic integration formulas

1. $\int k\,dx = kx + C$ (any number k)

2. $\int x^n\,dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$

3. $\int \frac{dx}{x} = \ln|x| + C$

4. $\int e^x\,dx = e^x + C$

5. $\int a^x\,dx = \frac{a^x}{\ln a} + C \quad (a > 0,\ a \neq 1)$

6. $\int \sin x\,dx = -\cos x + C$

7. $\int \cos x\,dx = \sin x + C$

8. $\int \sec^2 x\,dx = \tan x + C$

8. $\int \csc^2 x\,dx = -\cot x + C$

10. $\int \sec x \tan x\,dx = \sec x + C$

11. $\int \csc x \cot x\,dx = -\csc x + C$

12. $\int \tan x\,dx = \ln|\sec x| + C$

13. $\int \cot x\,dx = \ln|\sin x| + C$

14. $\int \sec x\,dx = \ln|\sec x + \tan x| + C$

15. $\int \csc x\,dx = -\ln|\csc x + \cot x| + C$

16. $\int \sinh x\,dx = \cosh x + C$

17. $\int \cosh x\,dx = \sinh x + C$

18. $\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1}\left(\frac{x}{a}\right) + C$

19. $\int \frac{dx}{a^2 + x^2} = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + C$

20. $\int \frac{dx}{x\sqrt{x^2 - a^2}} = \frac{1}{a}\sec^{-1}\left|\frac{x}{a}\right| + C$

21. $\int \frac{dx}{\sqrt{a^2 + x^2}} = \sinh^{-1}\left(\frac{x}{a}\right) + C \quad (a > 0)$

22. $\int \frac{dx}{\sqrt{x^2 - a^2}} = \cosh^{-1}\left(\frac{x}{a}\right) + C \quad (x > a > 0)$

Example 1: Making a Simplifying Substitution

$\int \frac{2x-9}{\sqrt{x^2-9x+1}} dx = 2\sqrt{x^2 - 9x + 1} + C$, by choosing $u = x^2 - 9x + 1$.

Example 2: Completing the Square $\int \frac{dx}{\sqrt{6x - x^2}} = \sin^{-1}\left(\frac{x-3}{3}\right) + C$

Example 3: Expanding a Power and Using a Trigonometric Identity

$$\int (\sec x + \tan x)^2 dx = 2\tan x + 2\sec x - x + C$$

Example 4: Eliminating a Square Root

$$\int_0^{\frac{\pi}{4}} \sqrt{1 + \cos 4x}\, dx =$$

Example 5: Reducing an Improper Fraction

$$\int \frac{4x^2+3}{2x-1}\,dx =$$

Example 6: Separating a Fraction

$$\int \frac{1-4x}{x^2+9} dx =$$

Extra Examples:

1. $\int \frac{dx}{\sqrt{8-e^{2x}}}$

2. $\int_{1}^{e} \frac{dx}{x\sqrt{4+(\ln x)^2}}$

3. $\int_{0}^{\ln\sqrt{3}} \frac{e^x}{1+e^{2x}}\,dx$

4. $\int_{\sqrt{2}}^{3} \frac{2x^3}{x^2-1}\,dx$

5. $\int_{0}^{\pi} \sqrt{1-\cos 4x}\,dx$

6. $\int \frac{dx}{1+\sin x}$